

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Summer Examination 2023

Course: B. Tech. **Branch :** Artificial Intelligence and Data Science (& All Allied Branches)

Subject Code & Name: BTAIC502 - Machine Learning

Semester : V

Max Marks: 60

Date: 09.08.2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		(12)
A) Explain Supervised Learning technique with example?	(Remember)	6
B) Explain Reinforcement Learning technique with example?	(Apply)	6
C) Explain several Issues in Machine Learning?	(Understand)	6
Q.2 Solve Any Two of the following.		(12)
A) Explain Area Under ROC curve with an example?	(Create)	6
B) Explain R-Squared in regression model?	(Evaluate)	6
C) Explain Classification Accuracy with an example?	(Understand)	6
Q. 3 Solve Any Two of the following.		(12)
A) Explain how you can represent multi-variate regression model?	(Understand)	6
B) Explain Linear regression with an example?	(Analyze)	6
C) Write implementation of Logistic Regression from Sklearn?	(Remember)	6
Q.4 Solve Any Two of the following.		(12)
A) Explain ID3 decision tree algorithm?	(Remember)	6
B) Define and explain Entropy?	(Analyze)	6
C) Explain Data Bagging and Feature Selection in random forests?	(Remember)	6
Q. 5 Solve Any Two of the following.		(12)
A) Explain Naïve Bayes classifier with an example? Write its advantages and dis-advantages?	(Apply)	6
B) Explain k-Nearest Neighbors classification algorithm?	(Analyze)	6
C) Explain the implementation of SVM on Iris Using Sklearn?	(Apply)	6

*** End ***